

MADELEIN JORDAAN

Technical Engineer & AI Integration Specialist

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PROFESSIONAL SUMMARY

Highly adaptable, analytical, and execution-focused **Technical Engineer** specializing in **AI Integration, Workflow Automation, and Full-Stack Engineering**. Proven track record of bridging the gap between high-level business strategy and robust technical execution. Expert in designing structured multi-step AI agents, orchestrating complex BPMN 2.0 architectures, and optimizing high-performance frontend applications. Adept at leveraging modern developer ecosystems to eliminate operational bottlenecks, accelerate deployment times, and design intuitive, data-driven decision tools.

TECHNICAL ECOSYSTEM & COMPETENCIES

AI & Agentic Engineering

- LLM Orchestration & Schema Enforcement
- Context Window Management
- Autonomous Multi-step Agent Pipelines
- Claude Code, Gemini API & GPT ecosystems
- API tool-calling & dynamic fallback systems

Process Automation & Backend

- BPMN 2.0 (Flowable Orchestration)
- JavaScript (ES6+), Node.js, Python, C++
- GraphQL & RESTful API Architecture
- Relational Databases (Oracle SQL I-III, MySQL)
- DevOps (Linux VPS, Server Stack Setup)

Frontend & Graphics

- React, Redux, Tailwind CSS, TypeScript
- Material UI, Bootstrap
- 3D Graphics (WebGL, Three.js)
- Data Visualization (D3.js, Chart.js, Grafana)
- Desktop UI Layout (Qt Designer)

PROFESSIONAL EXPERIENCE

Senior Junior Front-End Developer / Business Process Engineer

Mi-C3 International | Malta

July 2024 – Present

- **UI Component Engineering:** Architect, implement, and maintain scalable user interface components using JavaScript, React, Redux, Material UI, HTML5, and CSS3.
- **API & Backend Integration:** Bridge complex enterprise web systems using Node.js, GraphQL integrations, and RESTful API endpoints.
- **Performance Optimization:** Actively profile front-end performance metrics, reducing bundle sizes, managing asynchronous states, and resolving rendering bottlenecks to ensure low latency.
- **Codebase Documentation:** Author internal technical documentation for frontend components, modular design patterns, and state-management schemas to optimize teammate onboarding.
- **Enterprise Version Control:** Manage code branches, continuous integration, and peer code reviews inside GitLab.

Junior Business Process Engineer

Mi-C3 International | Malta

January 2023 – June 2024

- **Application Orchestration:** Developed and shipped production applications by designing logical workflows using BPMN 2.0 / Flowable engines integrated via JavaScript, HTML, CSS, and GraphQL.
- **Enterprise Software Customization:** Customized core modules inside the company's proprietary platform, Affectli, ensuring tailored business architecture for corporate clients.
- **Data Analytics Dashboards:** Engineered live monitoring data visualizations using Grafana to showcase operational metrics to clients.
- **Technical Mentorship:** Assisted with the comprehensive training, code-reviewing, and operational onboarding of newly recruited Business Process Engineers.

Graduate Business Process Engineer

Mi-C3 International | Malta

August 2022 – December 2022

- Completed fast-tracked enterprise platform onboarding, writing backend JavaScript hooks and custom GraphQL data hooks to connect system models to client interfaces using BPMN workflows.

EDUCATION & CREDENTIALS

Bachelor of Science (BSc) in Computing

University of South Africa (UNISA)

Expected Graduation: November 2026

Rigorous, highly mathematical distance-learning computer science track maintained part-time concurrently alongside full-time international software engineering. **Currently finishing final two subjects.**

- **Software Engineering & Architecture:** Passed Advanced Object-Oriented Analysis, Data Structures (C++/Qt), Systems Development, Software Project Management, Human-Computer Interaction I & II.
- **Infrastructure, AI & Visual Math:** Passed Computer Graphics (WebGL/Three.js render pipelines), Advanced Database Design (Oracle SQL I, II, III), Computer Networks, Techniques of Artificial Intelligence, Linear Algebra, Formal Logic, and Theoretical Computer Science.

Professional Training & Prior Studies

- **Front-End Engineer Professional Certificate** | Codecademy (Completed Sept 2024): Comprehensive training in Algorithmic Analysis, Web Architecture, React State Lifecycle, and Advanced JS.
- **BCom Law (Commercial Law Track) Coursework** | UNISA (2014 – 2017): Passed 15 specialized modules (Commercial Law, Economics, Quantitative Modelling) before transitioning fully to computer science; provides solid enterprise compliance and risk mitigation perspective.

KEY ACHIEVEMENTS & LEADERSHIP

- **Grand Prize Winner – Petronas Fluid Art Movement:** Ranked top competitor across the African continent in an international digital engineering design project. Curated and selected for an official design tour in New York City and Montreal.
- **Official Government Vetting & Security Clearance:** Successfully finalized comprehensive background verification protocols, including polygraph testing, conducted by the South African State Security Agency (SSA).
- **International Logistics Volunteerism:** Deployed to remote areas in Malawi to direct logistical resource routing and nutritional planning under high infrastructural constraints for vulnerable youth centers.